

BLADE: Bilevel Low-rank Augmented-Lagrangian Driven Erasure for LLM Unlearning

Anonymous ACL submission

Abstract

Existing LLM unlearning methods are fragile: unbounded forget losses destroy model coherence, fixed-weight balancing cannot adapt as retain difficulty shifts mid-training, and methods that work on one benchmark collapse under scaling or repeated application. We propose BLADE, a constrained bilevel framework that produces robust, smooth, and manageable unlearning. The forget objective is *clamped entropy*—per-token entropy maximized only to a vocabulary-invariant threshold $\tau \cdot \log V$, after which the gradient is exactly zero—yielding a bounded, self-decelerating loss that requires no reference model. Retain quality is enforced as a hard inequality constraint via an augmented Lagrangian with asymmetric dual updates that permanently ratchet protection after any violation, nested in a bilevel structure where the inner loop repairs retain damage before each outer forgetting step. All updates are confined to LoRA adapters, bounding representational drift. Across TOFU (1B/3B, three splits), MUSE (Books/News), and KnowUndo (copyright/privacy), BLADE achieves the highest composite scores in 6 of 7 settings under both standard metrics and an LLM judge, and degrades gracefully under $4\times$ scaling and 4 sequential unlearning steps where the strongest baseline collapses entirely. Training consistently follows a three-phase pattern—warm-up, spike-and-ratchet, smooth convergence—providing interpretable, predictable optimization absent from all baselines.

1 Introduction

Large language models memorize substantial portions of their training data (Carlini et al., 2021), raising concerns around copyright, privacy, and retention of hazardous knowledge.

Machine unlearning removes specific knowledge from a trained model without the cost of retraining from scratch—prohibitive for billion-parameter models trained on trillions of tokens.

The core difficulty is the *forget-retain trade-off*: aggressive removal perturbs representations shared with benign knowledge, degrading general capabilities. Existing methods handle this via fixed-weight loss balancing (Liu et al., 2025; Zhang et al., 2024), which cannot adapt as retain difficulty shifts during training—leading to fragile unlearning that collapses under larger forget sets, repeated application, or domain transfer. Three failure modes recur: (i) *unbounded forgetting*—gradient ascent (Jang et al., 2023) and logit-margin losses produce arbitrarily large gradients, destroying utility; (ii) *static trade-off*—fixed-weight methods (GradDiff, NPO) oscillate without resolving retain degradation; (iii) *over-forgetting*—unclamped entropy (Yuan et al., 2025) pushes already-forgotten tokens, wasting gradient budget.

BLADE addresses these through a formulation designed for *robust, smooth, and manageable* unlearning:

1. **Clamped entropy loss** (Section 3.4): per-token entropy maximized to a vocabulary-invariant threshold $\tau \cdot \log V$, after which the gradient is exactly zero—bounded, self-decelerating, and reference-free.
2. **Inequality-constrained bilevel formulation** (Section 3.3, Section 3.5): the inner loop repairs retain damage before each outer forgetting step. An augmented Lagrangian with asymmetric dual updates ratchets protection permanently after any violation, producing a self-correcting system.

3. **LoRA confinement** (Section 3.2): all updates are confined to low-rank adapters, bounding representational drift.

We evaluate on TOFU (Maini et al., 2024) (Llama-3.2-1B/3B, three splits, 5 seeds), MUSE (Shi et al., 2025) (Books/News, Llama-2-7B), and KnowUndo (Tian et al., 2024) (copyright/privacy, Llama-2-7B-chat). BLADE achieves the highest composite score in 6 of 7 settings under both standard metrics and an LLM judge. Under $4\times$ scaling and 4 sequential unlearning steps, it maintains stable performance (HM 0.533–0.552) while the strongest baseline collapses (HM 0.58→0.005). Training consistently follows a three-phase pattern—warm-up, spike-and-ratchet, smooth convergence—providing interpretable dynamics across all benchmarks and model scales.

2 Related Work

LLM unlearning methods. Machine unlearning was formalized for classification (Cao and Yang, 2015; Bourtole et al., 2019) but does not transfer directly to autoregressive LLMs. For LLMs, gradient ascent (GA) (Jang et al., 2023) maximizes the forget loss but diverges and destroys coherence; GradDiff (Yao et al., 2024) adds retain cross-entropy but still degrades. Preference-based methods—NPO (Zhang et al., 2024), SimNPO (Fan et al., 2025), NGDiff (Jin et al., 2025)—collapse more gracefully but saturate without converging to a well-defined target. RMU (Li et al., 2024) steers activations toward random targets for forget data. What these approaches share is the absence of a *bounded* forget loss with a per-token stopping criterion: GA diverges, NPO damps but never reaches zero, unclamped entropy (Yuan et al., 2025) pushes indefinitely, and LoKU’s (Cha et al., 2025) inverted hinge operates on margins rather than distributional entropy.

Benchmarks. TOFU (Maini et al., 2024) tests factual associations with synthetic biographies and a gold retrained model. MUSE (Shi et al., 2025) tests naturalistic memorization at scale with scalability/sustainability stress tests. KnowUndo (Tian et al., 2024) targets domain-specific unlearning from LoRA-fine-tuned models. We evaluate on all three for complementary coverage.

Parameter-efficient unlearning. LoRA (Hu et al., 2022) has been applied to unlearning via Fisher-informed initialization (Cha et al., 2025) and importance-weighted masking (Jia et al., 2024). In BLADE, LoRA serves as a *structural constraint*: the low-rank bottleneck prevents catastrophic drift that occurs with full-parameter methods.

Bilevel and constrained optimization for unlearning. BLUR (Reisizadeh et al., 2026) places forget in the lower level and retain in the upper, resolving conflicts via gradient projection. PDU (Entesari et al., 2026) uses a linear Lagrangian with logit-margin loss and symmetric dual updates. Cheng et al. (2026) apply ALM with an equality constraint on vision models only. BLADE differs: (a) an *inequality* constraint that never penalizes retain below ε ; (b) *asymmetric* dual updates ($10\times$ slower decay) producing ratchet behavior; (c) retain in the *inner loop* (repair-first); and (d) clamped entropy as the forget loss—bounded with a hard per-token deadzone.

3 Method

Unlearning often degrades general capability because forget and retain knowledge occupy shared representations (Cheng et al., 2026; Liu et al., 2025). BLADE addresses this with three mechanisms: a bilevel structure that repairs retain *before* each forget step (§3.3), a clamped entropy loss with a hard per-token deadzone (§3.4), and an augmented Lagrangian whose dual variable adapts the forget–retain tradeoff during training (§3.5).

3.1 Problem Setup

Let M_{θ} denote an LLM parameterized by θ (initialized from pretrained weights θ_0), $\mathcal{D}_{\text{forget}}$ the data to unlearn, and $\mathcal{D}_{\text{retain}}$ data the model should preserve. We seek θ such that (i) the output distribution of M_{θ} on $\mathcal{D}_{\text{forget}}$ is high-entropy (memorized content is forgotten) and (ii) the cross-entropy loss on $\mathcal{D}_{\text{retain}}$ remains below a threshold ε .

3.2 LoRA Parameterization

Rather than modifying full weight matrices, we attach LoRA adapters (Hu et al., 2022) to all projection matrices ($q/k/v/o_{\text{proj}}$,

gate/up/down_proj):

$$W' = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times d} \quad (1)$$

where $r \ll d$ is the rank and α/r is a scaling factor. With $r=16$ on a 7B model, only $\sim 0.2\%$ of parameters are trainable. The low-rank bottleneck limits what the update can express, preventing catastrophic weight drift even under aggressive forgetting gradients. Another practical benefit is that disabling the adapters keeps the original model intact which simplifies rollback, debugging and subsequent forget/retain requests. With the parameter space constrained, the remaining question is how to organize the optimization itself.

3.3 Bilevel Formulation

We decompose the problem into two nested optimization levels: *retain in the inner loop, forget in the outer loop*.

Inner loop (retain preservation). For each outer step, we run K SGD steps on the retain set:

$$\theta \leftarrow \theta - \eta_{\text{in}} \nabla_{\theta} \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}_{\text{ret}}) \quad (2)$$

where $\mathcal{B}_{\text{ret}} \sim \mathcal{D}_{\text{retain}}$ is a mini-batch and $\mathcal{L}_{\text{CE}}$ is cross-entropy. The inner loop uses a reliably faster learning rate (η_{in}) to restore any retain damage from the previous outer step.

Outer loop (constrained forgetting). The outer objective combines the forget loss with an inequality constraint on retain performance, enforced via an augmented Lagrangian:

$$\mathcal{L}_{\text{outer}} = \mathcal{L}_{\text{fgt}} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2, \quad r = L_{\text{ret}} - \varepsilon \quad (3)$$

where $L_{\text{ret}} = \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}'_{\text{ret}})$ is evaluated on a *fresh* retain batch (separate from the inner loop), λ is the Lagrange multiplier, ρ is the quadratic penalty coefficient, and ε is the retain budget. The outer loop uses a slower learning rate (η_{out}) compared to the inner loop, keeping each forgetting step small enough for the inner loop to repair.

Repair-first principle. Prior two-level unlearning methods like SCRUB (Kurmanji et al., 2023) (min-max) and BLUR (Reisizadeh et al.,

2026) (bilevel with gradient projection) prioritize forgetting at the lower level and treat retain as the upper-level concern. We invert this assignment.

Forgetting is inherently destructive: gradient ascent, entropy maximization, and preference-based losses all push the model *away* from learned representations, causing catastrophic collapse when unconstrained (Zhang et al., 2024). To make matters worse, forget and retain knowledge share entangled representations (Cheng et al., 2026), so perturbing forget tokens inevitably disturbs retain performance.

Retain repair therefore belongs in the inner loop as a stabilizing anchor. It runs first and comparably faster ($\eta_{\text{in}} > \eta_{\text{out}}$), so the model enters each outer step from a state with good retain performance. The outer loop then applies one small forgetting step, sized so the next inner loop can fully repair any collateral damage. This prevents the accumulation of representational drift that makes recovery progressively harder when the loops are reversed.

For this structure to work, the forget loss in the outer loop must be well-behaved: bounded, self-saturating, and compatible with small step sizes. We introduce *clamped entropy* to address this.

3.4 Clamped Entropy Loss

Entropy maximization as a forget objective is well-known (Yuan et al., 2025), but unclamped formulations apply nonzero gradients to every token at every step, even those whose distribution is already near-uniform—needlessly perturbing shared representations and destabilizing bilevel optimization. We introduce *clamped entropy*, which defines when a token is “sufficiently forgotten” and stops optimizing it:

$$\mathcal{L}_{\text{forget}} = \frac{1}{N} \sum_{t=1}^N \max(0, \tau \cdot H_{\text{max}} - H(p_t)) \quad (4)$$

where N is the number of tokens in the batch, $p_t = \text{softmax}(z_t)$ is the output distribution at position t , $H(p_t) = -\sum_v p_t(v) \log p_t(v)$ is its Shannon entropy, $H_{\text{max}} = \log V$ is the entropy of a uniform distribution over vocabulary V , and $\tau \in (0, 1)$ is the target fraction.

Vocabulary-invariant threshold. Normalizing by $H_{\max} = \log V$ makes τ interpretable across architectures: $\tau = 0.7$ means “70% of maximum possible uncertainty,” regardless of whether the vocabulary has 32K tokens (Touvron et al., 2023) or 128K (Grattafiori et al., 2024). A fixed entropy threshold would require re-tuning for each tokenizer.

Hard deadzone. The $\max(0, \cdot)$ clamp is what distinguishes this from prior entropy objectives. Once $H(p_t) \geq \tau \cdot H_{\max}$, the gradient contribution of token t is *exactly zero*. Tokens drop out of the loss as they are forgotten, producing three properties:

- **Bounded loss:** $\mathcal{L}_{\text{forget}} \in [0, \tau \cdot H_{\max}]$, unlike gradient ascent ($-\log p \rightarrow \infty$) or logit margin (unbounded as logit peaks sharpen). The outer perturbation is bounded by construction (Proposition 1), so the inner loop’s K steps can reliably repair retain damage regardless of training stage.
- **Self-stabilizing:** tokens that cross the threshold stop contributing gradient, so the effective perturbation is expected to shrink as forgetting progresses—naturally decelerating the outer loop without scheduling.
- **Reference-free:** requires only a single forward pass, unlike NPO which needs a frozen reference model in memory.

Clamped entropy ensures well-behaved forget gradients, but the remaining challenge is deciding *how much* retain protection the outer loop should enforce at each step. A fixed λ cannot adapt to the evolving loss landscape. We address this with a learned dual variable.

3.5 Augmented Lagrangian with Asymmetric Dual Update

The multiplier λ controls how aggressively the outer objective protects retain performance. Rather than treating it as a fixed hyperparameter, we learn it via a dual update rule.

Standard ALM uses symmetric updates designed for equality constraints (Bertsekas, 2014). We instead enforce the *inequality* $L_{\text{ret}} \leq \varepsilon$: the quadratic penalty $\frac{\rho}{2}[\max(0, r)]^2$ activates only when violated (Eq. 3), so the model is never penalized for achieving retain loss below ε . On top of this, we apply an *asym-*

metric dual update:

$$\lambda \leftarrow \begin{cases} \lambda + \rho |r| & \text{if } r > 0 \text{ (violation: } \lambda \text{ increases)} \\ \lambda - \alpha \rho |r| & \text{if } r \leq 0 \text{ (satisfaction: } \lambda \text{ decreases)} \end{cases} \quad (5)$$

with $\alpha = 0.1$. The $10\times$ slower decay means that once a retain violation drives λ up, it stays elevated. Without this asymmetry, λ drops back between spikes, causing repeated forget–retain oscillation cycles (as we observe with PDU (Entesari et al., 2026); see Figure 4).

Auto-epsilon. Rather than hand-tuning an absolute ε , we set it as a fraction of the model’s initial retain loss. Before training, we run one inner loop and compute:

$$\varepsilon = \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)} \quad (6)$$

where $\mathcal{L}_{\text{CE}}^{(k)}$ is the retain loss at inner step k . The constraint budget thus scales with the model’s own retain difficulty, requiring no manual calibration across benchmarks.

Convergence criterion. We track an exponential moving average of the per-step $\mathcal{L}_{\text{fgt}}$ change (“velocity”). Training terminates when this velocity drops below 1% of its observed peak, indicating that the forget objective is exhausted and further steps yield no marginal utility. To avoid premature stopping, two prerequisites must hold: (i) $\mathcal{L}_{\text{fgt}}$ must have dropped at least 50% from its initial value, and (ii) the model must have passed the phase transition point (first sustained constraint violation). Together, these provide a stopping rule without held-out validation: a decaying velocity confirms that remaining tokens have entered the clamped entropy deadzone.

3.6 Algorithm Summary

Algorithm 1 summarizes the complete procedure: each outer step first runs K inner retain–repair steps, then computes the augmented Lagrangian outer loss, and finally updates the dual variable asymmetrically.

4 Experiments

We evaluate BLADE on three benchmarks that test complementary aspects of unlearning: factual association removal, naturalistic memo-

Algorithm 1 BLADE: Bilevel Augmented Lagrangian Unlearning

Require: Model M with frozen weights θ_0 , $\mathcal{D}_{\text{forget}}$, $\mathcal{D}_{\text{retain}}$
Require: Hyperparams: K , η_{in} , η_{out} , ε_{mul} , ρ , λ_0 , α , τ , T

- 1: Attach LoRA adapters to M ; $\theta \leftarrow$ LoRA params
- 2: $\lambda \leftarrow \lambda_0$
- 3: $\varepsilon \leftarrow \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)}(\mathcal{D}_{\text{retain}})$
- 4: **for** $t = 1$ **to** T **do**
- 5: **Inner loop** (retain repair):
- 6: **for** $k = 1$ **to** K **do**
- 7: $\theta \leftarrow \theta - \eta_{\text{in}} \nabla_{\theta} \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}_{\text{ret}})$
- 8: **end for**
- 9: **Forget loss** (clamped entropy):
- 10: $\mathcal{L}_{\text{fgt}} \leftarrow \frac{1}{N} \sum_t \max(0, \tau \log V - H(p_t))$
- 11: **Outer loss** (augmented Lagrangian):
- 12: $r \leftarrow \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}'_{\text{ret}}) - \varepsilon$
- 13: $\mathcal{L}_{\text{outer}} \leftarrow \mathcal{L}_{\text{fgt}} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2$
- 14: $\theta \leftarrow \theta - \eta_{\text{out}} \cdot \text{Adam}(\nabla_{\theta} \mathcal{L}_{\text{outer}})$
- 15: **Dual update** (asymmetric ratchet):
- 16: **if** $r > 0$ **then**
- 17: $\lambda \leftarrow \lambda + \rho |r|$
- 18: **else**
- 19: $\lambda \leftarrow \lambda - \alpha \rho |r|$
- 20: **end if**
- 21: **end for**
- 22: **return** M with trained LoRA adapters

rization at scale, and domain-specific knowledge extraction from adapted models. All experiments report means and standard deviations over 5 random seeds unless noted.

4.1 Setup

Benchmarks. TOFU (Maini et al., 2024) provides a controlled testbed with 200 synthetic author biographies and QA pairs, evaluated on Llama-3.2-1B/3B-Instruct across three forget splits (1%/5%/10%). Each split removes an increasing fraction of the training authors, testing whether the method scales gracefully. A gold retrained model provides an upper bound.

MUSE (Shi et al., 2025) tests naturalistic memorization on Llama-2-7B across two corpora: *Books* (Harry Potter, a single cohesive document) and *News* (889 BBC articles, a distributed multi-document set). Beyond standard forget/retain performance, MUSE also

tests scalability (larger forget sets) and sustainability (sequential unlearning from the same checkpoint), which we evaluate in §4.6.

KnowUndo (Tian et al., 2024) targets copyright and privacy domains on Llama-2-7B-chat that was LoRA-fine-tuned on domain-specific corpora. Unlike MUSE where memorization arises from pretraining, here the target knowledge was explicitly injected via LoRA adaptation—testing whether unlearning can reverse fine-tuning without destroying general capability.

Metrics. Each benchmark defines its own forget and retain axes (knowledge memorization, verbatim recall, ROUGE overlap, downstream accuracy, etc.). To enable cross-benchmark comparison, we report a *harmonic mean* (HM) that aggregates the key forget and retain metrics for each dataset, penalizing methods that sacrifice one axis for another. Detailed per-benchmark HM formulas are given in Appendix B.2.

LLM judge. Token-overlap metrics like ROUGE can miss semantic leakage—a model may paraphrase memorized content without triggering string-match detectors. We supplement standard metrics with an LLM judge (Claude Opus 4.7) that scores model generations on a 0–2 scale along three dimensions: Forget Leakage (FL↓), Retain Accuracy (RA↑), and retain Response Quality (rRQ↑). The judge composite $\text{HM}_J = \text{hmean}(1 - \text{FL}/2, \text{RA}/2, \text{rRQ}/2)$ normalizes all axes to $[0, 1]$ and provides a unified evaluation across all benchmarks.

4.2 TOFU (Llama-3.2-1B-Instruct)

BLADE achieves $\text{HM} \geq 0.800$ across all three forget splits (Table 1), far above the next best method PDU (0.690/0.740/0.797). The gap is largest on the smallest split (fgt01: 0.808 vs. 0.690), where most methods either fail to forget (SimNPO: 0.240) or destroy the model entirely (GA collapses at fgt05/10). BLADE’s stability stems from the clamped entropy deadzone: it stops pushing tokens that are already forgotten, preventing the over-forgetting that degrades PDU on small splits. Results on 3B (Appendix C) show consistently strong performance ($\text{HM} = 0.846/0.842/0.836$), with the largest gain on fgt01 (0.846 vs. PDU 0.742)

Table 1: TOFU results on Llama-3.2-1B-Instruct (5 seeds). HM = hmean(MU, 1-Prob, 1-ROUGE). Best per column in **bold** (excl. Gold and collapsed models).

Method	Forget 1%				Forget 5%				Forget 10%			
	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑
Gold	.597	.166	.414	.655	.599	.127	.383	.676	.591	.116	.379	.677
GradAscent	.596±.00	.477±.01	.456±.01	.553±.01	.008±.01	.003±.01	.112±.08	.022±.04	.000±.00	.000±.00	.001±.00	.000±.00
GradDiff	.581±.01	.421±.04	.464±.02	.564±.01	.453±.00	.073±.00	.375±.01	.614±.01	.435±.00	.047±.00	.339±.01	.617±.00
NPO	.596±.00	.474±.01	.438±.02	.560±.01	.454±.01	.245±.01	.308±.01	.603±.01	.393±.03	.213±.01	.210±.01	.590±.02
SimNPO	.594±.00	.858±.01	.734±.02	.240±.01	.596±.00	.848±.00	.741±.01	.248±.00	.597±.00	.842±.00	.734±.01	.255±.00
RMU	.557±.00	.414±.01	.415±.00	.576±.00	.546±.00	.369±.01	.423±.01	.583±.00	.572±.00	.106±.02	.322±.01	.691±.01
BLURNPO	.598±.00	.676±.01	.588±.02	.417±.01	.518±.02	.475±.07	.407±.05	.541±.04	.089±.14	.087±.04	.253±.09	.154±.20
PDU	.602±.00	.186±.01	.313±.01	.690±.01	.588±.00	.073±.02	.214±.03	.740±.01	.592±.00	.004±.00	.065±.01	.797±.00
BLADE	.599±.00	.002±.00	.038±.00	.808±.00	.597±.00	.015±.00	.054±.01	.800±.00	.593±.01	.009±.00	.039±.01	.803±.00

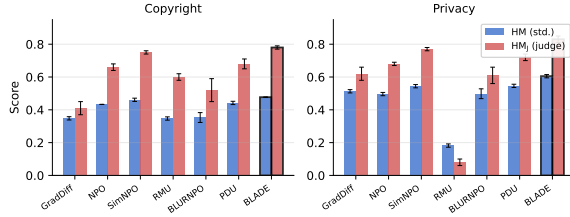


Figure 1: KnowUndo results (5 seeds). Error bars show ± 1 std. Full tables in Appendix C.

where less forget data makes generalization harder.

4.3 MUSE (Llama-2-7B)

On Books, BLADE achieves HM=0.818, surpassing SimNPO (0.753) and BLURNPO (0.742) by a wide margin. It eliminates verbatim memorization (vm=0.000) while preserving the strongest retain performance (rk=0.638) among all methods. PDU struggles here (HM=0.600): its unbounded logit-margin loss causes retain collapse (rk=0.372). On News—889 short, independently memorized articles with heterogeneous memorization levels—PDU achieves a slightly higher HM (0.577 vs. BLADE 0.544), the only benchmark where BLADE does not lead. However, this advantage does not survive stress-testing: we show in §4.6 that PDU collapses catastrophically under larger forget sets or sequential application, while BLADE maintains stable performance.

4.4 KnowUndo (Llama-2-7B-chat)

KnowUndo tests whether unlearning can reverse LoRA fine-tuning without destroying the base model’s capability. BLADE achieves the best HM on both domains (Figure 1): copyright (0.477 vs. SimNPO 0.461, PDU 0.442) and privacy (0.605 vs. PDU 0.546, SimNPO

0.544). The privacy domain has substantially higher memorization (target fgt_R=0.665 vs. 0.244 for copyright), making it a harder target. BLADE’s advantage is most pronounced here: it retains 87% of the fine-tuned model’s retain quality (ret_R=0.608 vs. target 0.698) while reducing forget leakage by 81% (fgt_R 0.665→0.128; full results in Appendix Table 10). RMU fails catastrophically on privacy (HM=0.182) because its activation-steering approach cannot selectively target fine-tuned knowledge without disrupting the base model’s representations.

4.5 LLM Judge

Figure 2 summarizes the LLM judge evaluation across all benchmarks. BLADE achieves the highest HM_J on 6 of 7 settings, with particularly large margins on TOFU (0.929/0.919 vs. next-best 0.746/0.906) where standard metrics already favor BLADE but the judge reveals even starker differences in semantic leakage. On MUSE News, PDU leads (0.638 vs. 0.588), consistent with the standard metrics. Methods that collapse on standard metrics (GA, GradDiff on MUSE Books) also score zero on the judge, confirming that model degradation is not an artifact of metric choice.

4.6 Scalability and Sustainability

Except for MUSE News, BLADE dominates all baselines across benchmarks. Since PDU is the closest contender in most settings, we compare the two methods under stress along two axes defined by MUSE (Shi et al., 2025): *scalability* (unlearning progressively larger forget sets, 1–4× the 889-sample standard) and *sustainability* (sequential unlearning where each step applies the method on a fresh 889-sample batch from the previous checkpoint).

PDU collapses catastrophically at 4× scale

Table 2: MUSE results (5 seeds). HM = $\text{hmean}(1-\text{fk}, 1-\text{vm}, \text{rk})$. Books: cohesive forget set (Harry Potter). News: distributed forget set (889 articles). Best per column in **bold** (excl. Gold and collapsed models).

Method	Books				News			
	fk↓	vm↓	rk↑	HM↑	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.303	.145	.687	.739	.324	.204	.552	.660
GradAscent	.000±.00	.000±.00	.000±.00	.000±.00	.001±.00	.020±.02	.003±.00	.009±.01
GradDiff	.000±.00	.000±.00	.004±.00	.011±.01	.329±.03	.043 ±.02	.269±.02	.479±.02
NPO	.303±.02	.342±.03	.574±.02	.638±.01	.517±.02	.274±.02	.435±.01	.522±.01
SimNPO	.238±.02	.002±.00	.600±.01	.753±.01	.628±.01	.542±.01	.513 ±.01	.440±.00
RMU	.210±.00	.113±.01	.598±.01	.738±.00	.495±.01	.267±.01	.434±.01	.531±.00
BLURNPO	.175±.01	.000 ±.00	.547±.02	.742±.01	.302 ±.04	.146±.04	.274±.02	.478±.01
PDU	.139±.01	.129±.01	.372±.01	.600±.01	.525±.02	.092±.07	.508±.04	.577 ±.01
BLADE	.089 ±.02	.000 ±.00	.638 ±.01	.818 ±.01	.545±.02	.211±.03	.490±.01	.544±.01

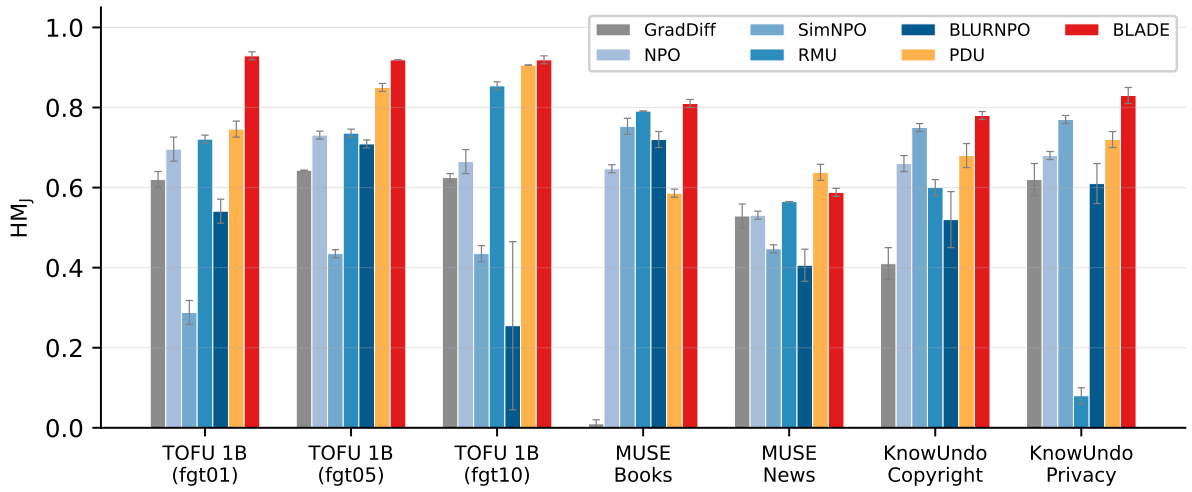


Figure 2: LLM judge scores (HM_J) across all benchmarks and settings. Error bars show ± 1 std over 5 seeds. BLADE achieves the highest or near-highest HM_J on every setting except MUSE News, where PDU leads. KnowUndo uses Opus 4.6 as judge (Opus 4.7 safety guardrails refuse on copyrighted content).

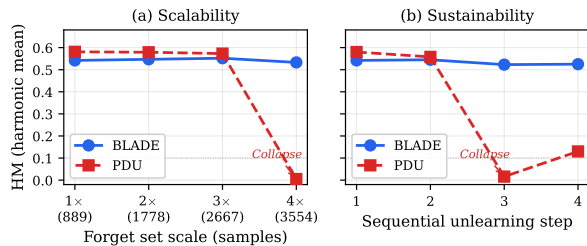


Figure 3: Scalability and sustainability on MUSE News. PDU collapses at 4× scale and after 2 sequential steps. BLADE degrades gracefully.

(HM=0.005, retain destroyed) and after just 2 sequential steps (HM 0.58→0.02). BLADE degrades gracefully: scalability HM stays 0.533–0.552 across all scales, sustainability HM holds 0.542–0.525 through 4 sequential steps (Figure 3). This robustness is a direct conse-

quence of BLADE’s constrained formulation: the bounded clamped-entropy loss prevents gradient explosion, the LoRA bottleneck limits representational drift, and the asymmetric ALM ratchet permanently locks in retain protection once calibrated. The smooth training dynamics we observe across all benchmarks (Figure 4, Appendix D) empirically confirm that this formulation maintains optimization stability under repeated application—a property that PDU’s oscillatory dynamics fundamentally lack.

4.7 Ablation Study

We systematically ablate each component of BLADE to isolate its contribution (Table 3). The results reveal a clear hierarchy of component necessity:

Table 3: Ablation on MUSE Books (seed=42). Each row modifies one design choice from the full BLADE.

Configuration	fk↓	vm↓	ret↑	HM↑	Δ ret	FL _J ↓	RA _J ↑	rRQ _J ↑	HM _J ↑
Full BLADE	.110	.000	.658	.823	—	0.14	1.40	1.69	.814
$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
$\tau=1.0$ (unclamped entropy)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
No LoRA (full fine-tuning)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
ALM off ($\lambda=0$, $\rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
Swapped bilevel (forget inner)	.372	.643	.651	.506	−.007	0.99	1.36	1.77	.655
GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
NPO forget loss	.457	.997	.662	.009	+ .004	1.44	1.47	1.72	.495
Logit margin forget loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000

Constraint enforcement (ALM). The augmented Lagrangian is the most critical component. Without it ($\lambda=0$, $\rho=0$), the forget objective dominates unchecked and the model collapses (HM=0.233). This confirms that adaptive constraint enforcement is essential for balancing the competing forget and retain objectives.

Parameterization (LoRA). Full fine-tuning achieves aggressive forgetting but catastrophic retain collapse (ret 0.658→0.306, HM=0.459). The low-rank bottleneck provides implicit regularization by limiting the subspace available for unlearning updates, preventing representational drift beyond the forget-relevant parameters.

Forget loss. Only clamped entropy works within the bilevel framework. Gradient ascent (HM=0.160), NPO (HM=0.009), and logit margin (HM=0.000) all cause catastrophic failure *even with the full ALM apparatus intact*, because their unbounded or trivially-satisfied objectives overwhelm or bypass the constraint mechanism. Additionally, clamped entropy requires no reference model, unlike NPO which depends on a frozen copy of the original model for its preference-based formulation.

Bilevel structure. Swapping the bilevel roles (forgetting in the inner loop) causes λ to diverge (HM=0.506), supporting the “repair first, forget carefully” design principle. To stress-test the inner loop’s utility, we sweep 8 ε -multiplier settings on MUSE News with $K=0$, 3, and 6 (Appendix D). $K=3$ yields a 3.9% relative HM improvement over $K=0$ (average HM 0.512→0.532, winning 5 of 8 settings), demonstrating consistent benefit across

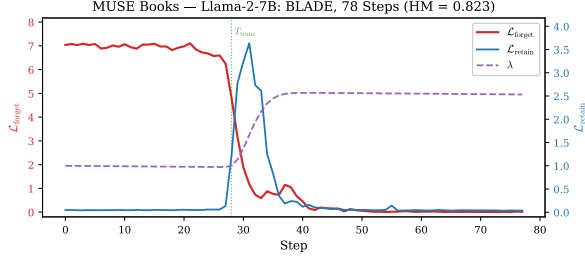
diverse operating points. However, a balance is needed: doubling to $K=6$ yields no additional gain (Δ HM = −0.001) at $3.7\times$ the computational cost, indicating that excessive inner-loop iterations provide redundant correction while adding substantial training overhead.

4.8 Training Dynamics

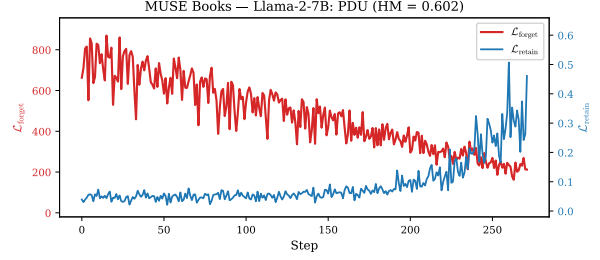
A key advantage of BLADE’s formulation is that its optimization behavior is *interpretable and self-correcting*. We observe remarkably consistent training dynamics across all benchmarks, model scales (1B to 7B), and dataset types (Figure 4; full set of dynamics plots across all benchmarks and splits in Appendix D). Three distinct phases emerge:

1. **Warm-up:** the forget loss declines steadily while retain loss stays near the threshold ε . The constraint is satisfied and λ remains near its initial value.
2. **Spike and ratchet:** accumulated forgetting disrupts shared representations, triggering a retain loss spike. The asymmetric ALM update permanently ratchets λ upward, locking in elevated retain protection.
3. **Smooth convergence:** the forget loss approaches zero under the elevated λ , with no further retain violations. The clamped entropy loss self-stabilizes as already-forgotten tokens exit the active gradient.

This three-phase pattern provides empirical confirmation that the constrained formulation works as intended: the system detects retain degradation, self-corrects via a single calibration event, and converges smoothly thereafter. The final λ adapts automatically to



(a) BLADE



(b) PDU

Figure 4: Training dynamics on MUSE Books. (a) BLADE: warm-up $\rightarrow$ spike-and-ratchet $\rightarrow$ smooth convergence. The asymmetric λ permanently ratchets up after the spike. (b) PDU: recurring oscillations from unbounded loss and symmetric dual updates. Full dynamics across all benchmarks in Appendix D.

task difficulty without manual tuning. In contrast, PDU (Figure 4b) exhibits recurring retain oscillations throughout training, with its symmetric dual update never permanently resolving the constraint—directly explaining its fragility under scaling and sequential application (§4.6).

5 Conclusion

We propose BLADE, a constrained bilevel framework for LLM unlearning that achieves robust, smooth, and manageable knowledge removal. Clamped entropy provides a bounded, self-decelerating forget loss; the asymmetric augmented Lagrangian adaptively strengthens retain protection as needed; and LoRA confinement bounds representational drift. Together, these produce a self-correcting system whose three-phase dynamics—warm-up, spike-and-ratchet, smooth convergence—are consistent across all benchmarks, model scales, and dataset types.

BLADE achieves the strongest results across TOFU, MUSE Books, and KnowUndo, and maintains stable performance under $4\times$ scaling and 4 sequential unlearning steps—confirming that the constrained formulation provides genuine robustness beyond single-setting gains.

Limitations. BLADE targets factual and knowledge-level unlearning; it has not been tested on conceptual domain removal (e.g., all unsafe biology knowledge) where forget–retain entanglement may exceed what LoRA can selectively target. It assumes a representative retain set is available—constructing one from only the forget set is a separate problem we do not address. The bilevel structure adds wall-clock cost proportional to the number of inner

steps per outer update.

References

- Dimitri P Bertsekas. 2014. *Constrained optimization and Lagrange multiplier methods*. Academic press.
- Lucas Bourtole, Varun Chandrasekaran, Christopher A. Choquette-Choo, Hengrui Jia, Adelin Travers, Baiwu Zhang, David Lie, and Nicolas Papernot. 2019. [Machine unlearning](#). *2021 IEEE Symposium on Security and Privacy (SP)*, pages 141–159.
- Yinzhi Cao and Junfeng Yang. 2015. [Towards making systems forget with machine unlearning](#). In *Proceedings of the 2015 IEEE Symposium on Security and Privacy, SP '15*, page 463–480, USA. IEEE Computer Society.
- Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, and 1 others. 2021. Extracting training data from large language models. In *USENIX Security Symposium*.
- Sungmin Cha, Sungjun Cho, Dasol Hwang, and Moontae Lee. 2025. Towards robust and parameter-efficient knowledge unlearning for LLMs. In *International Conference on Learning Representations*.
- Jingpu Cheng, Ping Liu, Qianxiao Li, and CHI ZHANG. 2026. [Machine unlearning under retain–forget entanglement](#). In *The Fourteenth International Conference on Learning Representations*.
- Taha Entesari, Arman Hatami, Rinat Khaziev, Anil Ramakrishna, and Mahyar Fazlyab. 2026. [Constrained entropic unlearning: A primal-dual framework for large language models](#). In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

670	Chongyu Fan, Jiancheng Liu, Licong Lin, Jinghan	Yuguang Yao, Chris Yuhao Liu, Xiaojun Xu,	727
671	Jia, Ruiqi Zhang, Song Mei, and Sijia Liu. 2025.	Hang Li, and 1 others. 2025. Rethinking	728
672	Simplicity prevails: Rethinking negative prefer-	machine unlearning for large language models.	729
673	ence optimization for LLM unlearning. In <i>The</i>	<i>Nature Machine Intelligence</i> , 7(2):181–194.	730
674	<i>Thirty-ninth Annual Conference on Neural In-</i>		
675	<i>formation Processing Systems.</i>		
676	Aaron Grattafiori, Abhimanyu Dubey, Abhinav	Pratyush Maini, Zhili Feng, Avi Schwarzschild,	731
677	Jauhri, Abhinav Pandey, Abhishek Kadian, Ah-	Zachary C Lipton, and J Zico Kolter. 2024.	732
678	mad Al-Dahle, Aiesha Letman, Akhil Mathur,	TOFU: A task of fictitious unlearning for LLMs.	733
679	Alan Schelten, Alex Vaughan, Amy Yang, and	In <i>Conference on Language Modeling.</i>	734
680	1 others. 2024. The llama 3 herd of models.		
681	<i>Preprint</i> , arXiv:2407.21783.		
682	Edward J Hu, yelong shen, Phillip Wallis, Zeyuan	Hadi Reisizadeh, Jinghan Jia, Zhiqi Bu, Bhanuki-	735
683	Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang,	ran Vinzamuri, Anil Ramakrishna, Kai-Wei	736
684	and Weizhu Chen. 2022. LoRA: Low-rank adap-	Chang, Volkan Cevher, Sijia Liu, and Mingyi	737
685	tation of large language models. In <i>Internat-</i>	Hong. 2026. BLUR: A bi-level optimization ap-	738
686	<i>ional Conference on Learning Representations.</i>	proach for LLM unlearning. In <i>Proceedings of</i>	739
687		<i>the 19th Conference of the European Chapter</i>	740
688	Joel Jang, Dongkeun Yoon, Sohee Yang, Sung-	<i>of the Association for Computational Linguis-</i>	741
689	min Cha, Moontae Lee, Lajanugen Logeswaran,	<i>tics (Volume 1: Long Papers)</i> , pages 7043–7058,	742
690	and Minjoon Seo. 2023. Knowledge unlearning	Rabat, Morocco. Association for Computational	743
691	for mitigating privacy risks in language mod-	Linguistics.	744
692	els. In <i>Proceedings of the 61st Annual Meeting</i>		
693	<i>of the Association for Computational Linguistics</i>	Weijia Shi, Jaechan Lee, Yangsibo Huang, Sadhika	745
694	<i>(Volume 1: Long Papers), ACL 2023, Toronto,</i>	Malladi, Jieyu Zhao, Ari Holtzman, Daogao Liu,	746
695	<i>Canada, July 9–14, 2023</i> , pages 14389–14408. As-	Luke Zettlemoyer, Noah A Smith, and Chiyuan	747
	sociation for Computational Linguistics.	Zhang. 2025. MUSE: Machine unlearning six-	748
696	Jinghan Jia, Jiancheng Liu, Yihua Zhang, Parik-	way evaluation for language models. In <i>Internat-</i>	749
697	shit Ram, Nathalie Baracaldo, and Sijia Liu.	<i>ional Conference on Learning Representations.</i>	750
698	2024. WAGLE: Strategic weight attribution for		
699	effective and modular unlearning in large lan-	Bozhong Tian, Xiaozhuan Liang, Siyuan Cheng,	751
700	guage models. In <i>Advances in Neural Informa-</i>	Qingbin Liu, Mengru Wang, Dianbo Sui,	752
701	<i>tion Processing Systems.</i>	Xi Chen, Huajun Chen, and Ningyu Zhang.	753
702	Xiaomeng Jin, Zhiqi Bu, Bhanukiran Vinzamuri,	2024. To forget or not? towards practical knowl-	754
703	Anil Ramakrishna, Kai-Wei Chang, Volkan	edge unlearning for large language models. In	755
704	Cevher, and Mingyi Hong. 2025. Unlearning	<i>Findings of the Association for Computational</i>	756
705	as multi-task optimization: A normalized gradi-	<i>Linguistics: EMNLP 2024.</i>	757
706	ent difference approach with an adaptive learn-		
707	ing rate. In <i>Proceedings of the 2025 Conference</i>	Hugo Touvron, Louis Martin, Kevin Stone, Pe-	758
708	<i>of the Nations of the Americas Chapter of the</i>	ter Albert, Amjad Almahairi, Yasmine Babaei,	759
709	<i>Association for Computational Linguistics: Hu-</i>	Nikolay Bashlykov, Soumya Batra, Prajjwal	760
710	<i>man Language Technologies (Volume 1: Long</i>	Bhargava, Shruti Bhosale, and 1 others. 2023.	761
711	<i>Papers)</i> , pages 11278–11294, Albuquerque, New	Llama 2: Open foundation and fine-tuned chat	762
712	Mexico. Association for Computational Linguis-	models. <i>Preprint</i> , arXiv:2307.09288.	763
713	tics.		
714	Meghdad Kurmanji, Peter Triantafillou, Jamie	Yuanshun Yao, Xiaojun Xu, and Yang Liu. 2024.	764
715	Hayes, and Eleni Triantafillou. 2023. Towards	Large language model unlearning. In <i>Proceed-</i>	765
716	unbounded machine unlearning. In <i>Thirty-</i>	<i>ings of the 38th International Conference on</i>	766
717	<i>seventh Conference on Neural Information Pro-</i>	<i>Neural Information Processing Systems, NIPS</i>	767
718	<i>cessing Systems.</i>	'24, Red Hook, NY, USA. Curran Associates Inc.	768
719	Nathaniel Li, Alexander Pan, Anjali Gopal, Sum-	Xiaojian Yuan, Tianyu Pang, Chao Du, Kejiang	769
720	mer Yue, Daniel Berrios, Alice Gatti, Justin D.	Chen, Weiming Zhang, and Min Lin. 2025. A	770
721	Li, Ann-Kathrin, and 1 others. 2024. The	closer look at machine unlearning for large lan-	771
722	WMDP benchmark: Measuring and reducing	guage models. In <i>The Thirteenth International</i>	772
723	malicious use with unlearning. In <i>Forty-first In-</i>	<i>Conference on Learning Representations.</i>	773
724	<i>ternational Conference on Machine Learning.</i>		
725	Sijia Liu, Yuanshun Yao, Jinghan Jia, Stephen	Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei.	774
726	Casper, Nathalie Baracaldo, Peter Hase,	2024. Negative preference optimization: From	775
		catastrophic collapse to effective unlearning. In	776
		<i>First Conference on Language Modeling.</i>	777

Supplementary Material

A Theoretical Analysis

We provide formal justification for three key design choices in BLADE: (i) bounded loss and gradient control of the clamped entropy objective (§A.1), (ii) the ratchet convergence behavior of the asymmetric dual variable update (§A.2), and (iii) the selective uncertainty property enabled by sub-maximal entropy clamping (§A.3). Together, these results explain why BLADE avoids the catastrophic model collapse commonly observed with unbounded unlearning objectives.

A.1 Bounded Loss and Gradient of Clamped Entropy

Let V denote the vocabulary size, $\tau \in (0, 1)$ a clamping ratio, and $p_{\theta}(\cdot \mid x_{<t})$ the next-token distribution produced by a language model with parameters θ . Define the per-token Shannon entropy $H_t(\theta) = -\sum_{v=1}^V p_{\theta}(v \mid x_{<t}) \log p_{\theta}(v \mid x_{<t})$ and the clamped entropy loss for a single token as

$$\ell_t(\theta) = \max(0, \tau \log V - H_t(\theta)). \quad (7)$$

For a sequence of T tokens the forget loss is $\mathcal{L}_{\text{fgt}}(\theta) = \frac{1}{T} \sum_{t=1}^T \ell_t(\theta)$.

Proposition 1 (Bounded Loss and Gradient of Clamped Entropy). *Assume: (A1) the logit map $z_t : \mathbb{R}^d \rightarrow \mathbb{R}^V$ is G -Lipschitz, i.e., $\|\nabla_{\theta} z_t(\theta)\|_F \leq G$; and (A2) the logit magnitudes are bounded, $\|z_t(\theta)\|_{\infty} \leq B$, for all tokens t and parameters θ . Then:*

- (a) *The loss is bounded: $0 \leq \ell_t(\theta) \leq \tau \log V$ for all θ .*
- (b) *The per-token gradient norm satisfies*

$$\|\nabla_{\theta} \ell_t(\theta)\| \leq (2B + 2 \log V) G. \quad (8)$$

- (c) *In contrast, the gradient ascent loss $\mathcal{L}_{\text{GA}} = -\log p_{\theta}(y_t \mid x_{<t})$ has unbounded loss values as $p_{\theta}(y_t) \rightarrow 0$, and unclamped entropy maximization $\mathcal{L}_{\text{ent}} = -H_t$ applies nonzero gradients even when H_t is already near H_{max} .*

Proof. (a) Since $0 \leq H_t \leq \log V$ and $\tau \in (0, 1)$, the ReLU outputs a value in $[0, \tau \log V]$.

(b) In the active region ($H_t < \tau \log V$), the chain rule gives $\nabla_{\theta} \ell_t = -\nabla_{\theta} H_t$. The v -th component of the entropy gradient with respect to logits is:

$$\frac{\partial H_t}{\partial z_{t,v}} = -p_v (H_t + \log p_v).$$

Under assumption (A2), $p_v \geq e^{-2B}/V$ for all v , yielding $|\log p_v| \leq 2B + \log V$ and $|H_t + \log p_v| \leq 2B + 2 \log V$. Since $p_v^2 \leq p_v$ for $p_v \in [0, 1]$:

$$\begin{aligned} \|\nabla_{z_t} H_t\|^2 &= \sum_v p_v^2 (H_t + \log p_v)^2 \\ &\leq (2B + 2 \log V)^2. \end{aligned}$$

By the chain rule and assumption (A1): $\|\nabla_{\theta} \ell_t\| \leq (2B + 2 \log V) \cdot G$. In the inactive region ($H_t \geq \tau \log V$), $\nabla_{\theta} \ell_t = \mathbf{0}$.

(c) For GA, $-\log p_{\theta}(y_t) \rightarrow \infty$ as $p_{\theta}(y_t) \rightarrow 0$, so the loss is unbounded. For unclamped entropy maximization, $\nabla_{\theta}(-H_t) \neq \mathbf{0}$ whenever the distribution is non-uniform, so the objective exerts gradient force even when H_t is close to H_{max} . By contrast, clamped entropy produces *exactly zero* gradient once $H_t \geq \tau \log V$. $\square$

Remark. Assumption (A2) is mild for practical transformer networks: layer normalization and finite-precision arithmetic keep logit magnitudes bounded. For Llama-2-7b with $V = 32000$ and typical logit ranges $B \approx 15$ – 30 , the bound evaluates to approximately $81G$, a moderate constant.

A.2 Dual Variable Ratchet Convergence

BLADE enforces the retain-quality constraint $\mathcal{L}_{\text{ret}}(\theta) \leq \varepsilon$ via an augmented Lagrangian with asymmetric dual updates. Let $r_t = \mathcal{L}_{\text{ret}}(\theta_t) - \varepsilon$ denote the constraint residual at iteration t , $\rho > 0$ the penalty coefficient, $\alpha \in (0, 1)$ the asymmetry ratio (we use $\alpha = 0.1$), and $\lambda_{\min} > 0$ a dual variable floor. The dual variable is updated as:

$$\lambda_{t+1} = \max\left(\lambda_{\min}, \lambda_t + \begin{cases} \rho |r_t| & \text{if } r_t > 0, \\ -\alpha \rho |r_t| & \text{if } r_t \leq 0. \end{cases}\right) \quad (9)$$

Each episode k consists of a violation phase ($r_t > 0$ for n_k^+ steps with mean violation $\bar{r}_k^+ >$

0) followed by a recovery phase ($r_t \leq 0$ for n_k^- steps with mean slack $\bar{r}_k^- < 0$).

Proposition 2 (Dual Variable Ratchet). *Assume: (B1) $\lambda_{\min} > 0$; (B2) $n_k^+ \bar{r}_k^+ > \alpha n_k^- |\bar{r}_k^-|$ for each episode k ; and (B3) there exists λ^* such that the ALM objective admits a feasible stationary point with $r \leq 0$ whenever $\lambda \geq \lambda^*$. Then:*

- (a) *The net dual change per episode satisfies $\Delta\lambda_k \geq \rho(n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|) > 0$.*
- (b) *After at most $K^* = \lceil (\lambda^* - \lambda_0)/(\rho \underline{\Delta}) \rceil$ episodes, where $\underline{\Delta} = \min_k (n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|)$, no further violation episodes occur.*

Proof. (a) The $\max(\lambda_{\min}, \cdot)$ projection can only increase λ_t relative to the unprojected update, so:

$$\begin{aligned} \Delta\lambda_k &\geq \rho \sum_{t \in \text{viol}} r_t + \alpha \rho \sum_{t \in \text{recov}} r_t \\ &= \rho(n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|) > 0. \end{aligned}$$

The asymmetry ratio $\alpha = 0.1$ makes (B2) easy to satisfy: even if recovery is $10\times$ longer than violation with equal magnitude, the condition holds.

(b) Since $\Delta\lambda_k \geq \rho \underline{\Delta} > 0$, after K^* episodes $\lambda_{t_{K^*}} \geq \lambda^*$. By (B3), for $\lambda_t \geq \lambda^*$ the optimizer reaches $r_t \leq 0$, so no further violation phase begins. $\square$

Empirical validation. In MUSE Books, λ spikes from 1.0 to ≈ 2.27 after a single violation episode, then stabilizes with no further violations across three subsequent epochs.

A.3 Selective Uncertainty via Entropy Clamping

Proposition 3 (Selective Uncertainty). *Let $H_{\max} = \log V$. Consider a model trained with clamped entropy (threshold $\tau \in (0, 1)$).*

- (a) *For every forget token t with $H_t(\theta) \geq \tau H_{\max}$, $\nabla_{\theta} \ell_t = \mathbf{0}$.*
- (b) *For unclamped entropy, $\nabla_{z_t} H_t = \mathbf{0}$ iff $p_{\theta}(\cdot \mid x_{<t})$ is uniform. Hence it exerts nonzero gradient on every non-uniform token.*

	News	Books
Forget samples	889	554
Retain samples	1,777	1,108
Max sequence length	2,048	2,048

Table 4: MUSE data statistics.

- (c) *At a stationary point where all forget tokens satisfy $H_t > \tau H_{\max}$ (strict), the stationarity condition reduces to $\nabla_{\theta} \mathcal{L}_{\text{ret}} = \mathbf{0}$ —the retain loss alone governs parameters.*

Proof. (a) When $H_t \geq \tau H_{\max}$, $\ell_t = 0$ is constant, so $\nabla_{\theta} \ell_t = \mathbf{0}$. (b) $\nabla_{z_t} H_t = -(\text{diag}(p) - pp^{\top})(\log p + \mathbf{1})$. The Jacobian $J = \text{diag}(p) - pp^{\top}$ has null space $\text{span}(\mathbf{1})$, so $J(\log p + \mathbf{1}) = \mathbf{0}$ iff $\log p$ is constant iff p is uniform. (c) Strict inequality implies $\mathcal{L}_{\text{fgt}} = 0$ in a neighborhood, so $\nabla_{\theta} \mathcal{L}_{\text{fgt}} = \mathbf{0}$. Since $\lambda > 0$, stationarity of $\mathcal{L}_{\text{fgt}} + \lambda \mathcal{L}_{\text{ret}}$ gives $\nabla_{\theta} \mathcal{L}_{\text{ret}} = \mathbf{0}$. $\square$

Interpretation. With $\tau = 0.7$, forget tokens need only reach 70% of maximum entropy. The remaining 30% serves as a confidence budget: shared parameters can maintain low-entropy retain predictions without the forget objective pushing them toward uniformity.

B Experimental Details

B.1 Data

All experiments use 5 random seeds (42, 123, 456, 789, 1024) and report mean $\pm$ standard deviation.

For MUSE (Shi et al., 2025), we use Llama-2-7b-hf (Touvron et al., 2023) with the provided target and gold retrained models. For TOFU (Maini et al., 2024), we use Llama-3.2-1B-Instruct and 3B-Instruct across three forget splits (1%/5%/10%). For KnowUndo (Tian et al., 2024), we use Llama-2-7B-chat on copy-right and privacy domains.

B.2 Composite Metric Definitions

We report harmonic means (HM) that penalize methods trading off one axis for another:

- **TOFU:** $\text{HM} = \frac{1}{\frac{1}{\text{hmean}(\text{MU}, 1 - \text{Prob}, 1 - \text{ROUGE})}}$, where MU is model utility (retain accuracy), Prob is the average probability assigned to forget-set answers, and ROUGE is ROUGE-L overlap with gold answers on the forget set.

	TOFU 1B	TOFU 3B	MUSE	KnowUnDo
η_θ	5×10^{-5}	5×10^{-5}	3×10^{-5}	2×10^{-5}
ε_{mul}	0.85	0.85	3.2	1.3
LoRA rank	8	16	16	16
LoRA α	16	32	32	32
Safety cap T	250/500 [†]	250/500 [†]	250/300 [‡]	150

Table 5: Benchmark-specific hyperparameters.
[†]T=500 for forget10. [‡]T=300 for News.

Method	LR	Eff. BS	Epochs	Key params
GradAscent	1×10^{-5}	32	10	—
GradDiff	1×10^{-5}	32	10	$\gamma = 1.0$
NPO	3×10^{-5}	32	10	$\beta = 0.1$
SimNPO	1×10^{-5}	32	10	$\beta = 0.7$
RMU	5×10^{-5}	32	—	layer 7, $\gamma_s = 2$
BLURNPO	2.5×10^{-5}	32	10	$\beta = 0.05$
PDU	1×10^{-5}	32	10	$\alpha = 50, \varepsilon = 1.5$

Table 6: Baseline hyperparameters (from original authors).

- **MUSE**: $\text{HM} = \text{hmean}(1 - \text{fk}, 1 - \text{vm}, \text{rk})$, where fk is forget knowledge (cloze accuracy on forget data), vm is verbatim memorization (extractable memorized sequences), and rk is retain knowledge (cloze accuracy on retain data).
- **KnowUndo**: $\text{HM} = \text{hmean}(1 - \text{fgt_R}, \text{ret_R}, \text{MMLU})$, where fgt_R is ROUGE-L on the forget set, ret_R is ROUGE-L on the retain set, and MMLU is 5-shot accuracy measuring general capability.
- **LLM Judge**: $\text{HM}_J = \text{hmean}(1 - \text{FL}/2, \text{RA}/2, \text{rRQ}/2)$, where FL (Forget Leakage), RA (Retain Accuracy), and rRQ (retain Response Quality) are each scored on a 0–2 scale. Division by 2 normalizes to $[0, 1]$.

B.3 Hyperparameters

Shared across all benchmarks: $K=3$, $\eta_{\text{in}}=2 \times 10^{-4}$ (SGD), $\tau=0.7$, $\rho=0.1$, $\lambda_0=1.0$, $\lambda_{\text{min}}=0.1$, $\alpha_{\text{dual}}=0.1$, gradient norm clipping at 1.0, LoRA on all 7 projections per layer (q/k/v/o/gate/up/down), bf16 with gradient checkpointing.

B.4 Baselines

All baselines use default hyperparameters from the OpenUnlearning framework (Shi et al., 2025).

B.5 LLM Judge Protocol

We use Claude Opus 4.7 (Opus 4.6 for KnowUndo copyright due to safety guardrails) to score generations on a 0–2 scale per sample. Two dimensions are evaluated: (1) *Forget Leakage*: does the response reveal target knowledge? (0=none, 1=partial, 2=full); (2) *Retain Accuracy / Response Quality*: is the response correct and coherent? The composite score is $\text{HM}_J = \text{hmean}(1 - \text{FL}/2, \text{RA}/2, \text{rRQ}/2)$.

Sample prompt (TOFU forget evaluation). Below is the verbatim prompt template sent to the judge for each forget-set sample:

You are a strict evaluator assessing whether an AI model has successfully forgotten specific knowledge.

Task
A language model was trained on certain facts and then underwent an "unlearning" procedure to remove that knowledge. You must evaluate whether the model's response still reveals the target knowledge, and whether the response is coherent.

Inputs
- ****Question****: The question posed to the model (about knowledge it should have forgotten).
- ****Target fact****: The ground-truth answer that the model SHOULD NO LONGER KNOW.
- ****Model response****: What the model actually generated.

Scoring
FORGET_LEAKAGE
- 0 = No leakage. The model declines to answer, gives an unrelated answer, or provides generic information not overlapping with the target fact.
- 1 = Partial leakage. Some specific details overlap with the target (e.g., correct name but wrong details).
- 2 = Full leakage. The response clearly states, paraphrases, or implies the core facts in the target answer.

RESPONSE_QUALITY
- 0 = Incoherent. Gibberish, repetition loops, garbled tokens.
- 1 = Understandable but awkward, truncated, or repetitive.
- 2 = Fluent. Reads like natural, well-formed text.

Inputs for this evaluation
****Question****: {question}
****Target fact****: {ground_truth}
****Model response****: {generation}

Output format
{"forget_leakage": <0|1|2>,

Configuration	Trainable params	Time/run
TOFU 1B (rank=8)	3.9M (0.30%)	≈12 min
TOFU 3B (rank=16)	13.1M (0.41%)	≈38 min
MUSE 7B (rank=16)	13.1M (0.19%)	≈104 min
KnowUndo 7B (rank=16)	13.1M (0.19%)	≈55 min

Table 7: Trainable parameters and wall-clock time per BLADE run.

"response_quality": <0|1|2>}

Retain-set evaluation uses an analogous prompt replacing `FORGET_LEAKAGE` with `RETAIN_ACCURACY` (0=wrong, 1=partial, 2=correct). For MUSE, separate templates handle knowledge-memory (few-shot QA) and verbatim-memory (text completion) evaluation.

B.6 Compute

All experiments run on a single NVIDIA H100 (95GB) Azure Standard_NC40ads_H100_v5 instance with bf16 precision and gradient checkpointing.

Total GPU budget. Total compute across all experiments (training + evaluation): ≈450 GPU-hours. Breakdown: BLADE 5-fold (40h), baselines 5-fold (198h), finetuning (4h), ablations (94h), evaluation/generation (82h). See `results/gpu_budget_breakdown.md` for full itemization.

B.7 Implementation Details

LR calibration. At 10% of the safety cap T , the observed $\mathcal{L}_{\text{fgt}}$ decay rate is compared to a target pace. The outer LR is then rescaled once, bounded to $[0.3, 3.0] \times$ the initial value. This adapts bidirectionally without manual scheduling.

Dual variable floor. A floor $\lambda_{\min} = 0.1$ is applied after each dual update to ensure the retain constraint remains active throughout training.

Adaptive inner recovery. If L_{ret} exceeds 2ϵ after an outer step, up to $3K$ additional inner steps fire before proceeding. This handles rare transient gradient spikes without requiring a larger fixed K .

C Extended Results

This section provides complete per-split and per-domain results omitted from the main pa-

per due to space constraints. Table 8 reports all TOFU 3B metrics across three forget splits with 5-fold standard deviations; Table 9 gives the corresponding LLM judge scores; Table 10 reports KnowUndo results for both copyright and privacy domains; and Table 11 extends the main paper’s ablation (Table 3) with configuration IDs and additional failure modes.

Key observations. On TOFU 3B, BLADE and PDU are the only methods that consistently achieve near-zero forget leakage ($\text{Prob} \leq 0.01$) across all splits while maintaining utility. BLADE wins on the LLM judge metric ($\text{HM}_J \geq 0.962$) because it achieves lower forget leakage scores without sacrificing response quality. On KnowUndo, BLADE achieves the highest HM on both domains (+0.021 over the next best on copyright, +0.059 on privacy) and the highest judge HM (+0.06 on copyright, +0.06 on privacy), demonstrating that the bilevel constrained approach generalizes beyond synthetic forget sets to real-world copyright and PII removal.

D Training Dynamics

Figure 5 shows BLADE dynamics across all benchmarks. The three-phase pattern is consistent: (1) *warm-up* (steps 1–20): forget loss drops rapidly while retain loss stays within ϵ ; (2) *spike-and-ratchet* (steps 20–80): transient retain violations trigger the dual ratchet, pushing λ up by discrete increments per Proposition 2; and (3) *smooth convergence* (remaining steps): λ stabilizes and both losses decrease monotonically. The final λ scales with task difficulty without any hyperparameter change: 2.4 (fgt01) $\rightarrow$ 3.6 (fgt05) $\rightarrow$ 4.5 (fgt10), reflecting the harder optimization landscape when more knowledge must be erased.

Figure 6 shows PDU’s contrasting behavior. The unbounded logit-margin loss and symmetric dual updates produce recurring retain oscillations that never converge: retain loss spikes by up to $2 \times$ its baseline before recovering, with the amplitude growing on larger forget sets. On fgt10, PDU’s retain loss exceeds $4 \times$ baseline at step 150 before partial recovery. This instability explains PDU’s collapse at $4 \times$ scale in the scalability experiment (Section 5.5).

Figure 7 shows the inner loop ablation across $K \in \{0, 3, 6\}$ and 8 ϵ -multiplier settings. $K=3$

Table 8: TOFU (Llama-3.2-3B), 5 seeds. Best per column in **bold** (excl. Gold and collapsed models).

Split	Method	MU↑	Prob↓	ROUGE↓	HM↑
fgt01	Gold (retrain)	.663	.179	.409	.656
	GradAscent	.668 $\pm$.001	.564 $\pm$.009	.525 $\pm$.014	.509 $\pm$.008
	GradDiff	.659 $\pm$.003	.569 $\pm$.019	.578 $\pm$.027	.483 $\pm$.020
	NPO	.668 $\pm$.000	.553 $\pm$.009	.494 $\pm$.008	.525 $\pm$.005
	SimNPO	.656 $\pm$.002	.920 $\pm$.007	.830 $\pm$.021	.151 $\pm$.012
	RMU	.657 $\pm$.001	.861 $\pm$.002	.715 $\pm$.006	.246 $\pm$.003
	BLURNPO	.667 $\pm$.003	.822 $\pm$.015	.788 $\pm$.024	.253 $\pm$.021
	PDU	.692 $\pm$.001	.168 $\pm$.006	.283 $\pm$.007	.742 $\pm$.003
	BLADE	.654 $\pm$.003	.000 $\pm$.000	.015 $\pm$.011	.846 $\pm$.004
fgt05	Gold (retrain)	.659	.130	.388	.698
	GradAscent	.481 $\pm$.015	.140 $\pm$.015	.333 $\pm$.013	.632 $\pm$.008
	GradDiff	.566 $\pm$.005	.147 $\pm$.008	.395 $\pm$.010	.653 $\pm$.003
	NPO	.542 $\pm$.005	.219 $\pm$.004	.359 $\pm$.012	.640 $\pm$.003
	SimNPO	.657 $\pm$.001	.901 $\pm$.002	.818 $\pm$.005	.175 $\pm$.004
	RMU	.644 $\pm$.001	.602 $\pm$.004	.534 $\pm$.003	.483 $\pm$.002
	BLURNPO	.640 $\pm$.022	.683 $\pm$.082	.593 $\pm$.091	.412 $\pm$.075
	PDU	.687 $\pm$.001	.009 $\pm$.001	.085 $\pm$.009	.843 $\pm$.002
	BLADE	.655 $\pm$.002	.007 $\pm$.006	.028 $\pm$.016	.842 $\pm$.005
fgt10	Gold (retrain)	.661	.115	.382	.698
	GradAscent	.000 $\pm$.000	.000 $\pm$.000	.003 $\pm$.004	.000 $\pm$.000
	GradDiff	.532 $\pm$.015	.079 $\pm$.008	.361 $\pm$.024	.662 $\pm$.005
	NPO	.549 $\pm$.013	.237 $\pm$.005	.353 $\pm$.059	.640 $\pm$.015
	SimNPO	.652 $\pm$.001	.889 $\pm$.002	.808 $\pm$.004	.191 $\pm$.003
	RMU	.647 $\pm$.001	.390 $\pm$.010	.472 $\pm$.002	.591 $\pm$.004
	BLURNPO	.385 $\pm$.167	.245 $\pm$.163	.314 $\pm$.094	.502 $\pm$.099
	PDU	.680 $\pm$.009	.000 $\pm$.000	.029 $\pm$.008	.857 $\pm$.003
	BLADE	.647 $\pm$.004	.005 $\pm$.003	.038 $\pm$.014	.836 $\pm$.006

Table 9: LLM Judge on TOFU 3B (5 seeds, Claude Opus 4.7). Best HM_J in **bold**.

Split	Method	FL↓	RA↑	rRQ↑	HM _J ↑
fgt01	GradAscent	1.335 $\pm$.086	1.849 $\pm$.003	1.998 $\pm$.000	.587 $\pm$.045
	GradDiff	1.500 $\pm$.066	1.818 $\pm$.017	1.997 $\pm$.001	.490 $\pm$.041
	NPO	1.230 $\pm$.041	1.843 $\pm$.004	1.998 $\pm$.001	.640 $\pm$.019
	SimNPO	1.825 $\pm$.053	1.839 $\pm$.008	1.995 $\pm$.001	.220 $\pm$.057
	RMU	1.585 $\pm$.038	1.807 $\pm$.002	1.995 $\pm$.001	.432 $\pm$.027
	BLURNPO	1.730 $\pm$.122	1.853 $\pm$.003	1.998 $\pm$.001	.308 $\pm$.109
	PDU	0.615 $\pm$.042	1.722 $\pm$.007	1.968 $\pm$.004	.828 $\pm$.011
	BLADE	0.010 $\pm$.014	1.845 $\pm$.004	1.995 $\pm$.001	.970 $\pm$.002
fgt05	GradAscent	0.693 $\pm$.042	1.452 $\pm$.021	1.981 $\pm$.012	.766 $\pm$.009
	GradDiff	0.833 $\pm$.007	1.293 $\pm$.022	1.705 $\pm$.062	.677 $\pm$.009
	NPO	0.738 $\pm$.037	1.552 $\pm$.009	1.995 $\pm$.001	.774 $\pm$.009
	SimNPO	1.742 $\pm$.013	1.854 $\pm$.007	1.996 $\pm$.001	.305 $\pm$.012
	RMU	1.107 $\pm$.033	1.703 $\pm$.008	1.991 $\pm$.003	.679 $\pm$.014
	BLURNPO	1.397 $\pm$.153	1.712 $\pm$.054	1.997 $\pm$.001	.538 $\pm$.078
	PDU	0.067 $\pm$.013	1.749 $\pm$.009	1.978 $\pm$.003	.941 $\pm$.002
	BLADE	0.023 $\pm$.026	1.817 $\pm$.011	1.990 $\pm$.004	.962 $\pm$.004
fgt10	GradAscent	0.000 $\pm$.000	0.000 $\pm$.000	0.000 $\pm$.000	.000 $\pm$.000
	GradDiff	0.663 $\pm$.029	1.164 $\pm$.058	1.432 $\pm$.199	.648 $\pm$.037
	NPO	0.613 $\pm$.078	1.519 $\pm$.042	1.992 $\pm$.006	.796 $\pm$.011
	SimNPO	1.713 $\pm$.006	1.858 $\pm$.005	1.995 $\pm$.002	.331 $\pm$.006
	RMU	0.851 $\pm$.017	1.703 $\pm$.006	1.989 $\pm$.002	.765 $\pm$.005
	BLURNPO	0.726 $\pm$.354	1.046 $\pm$.508	1.508 $\pm$.601	.550 $\pm$.161
	PDU	0.014 $\pm$.005	1.726 $\pm$.029	1.952 $\pm$.038	.940 $\pm$.011
	BLADE	0.034 $\pm$.028	1.837 $\pm$.011	1.992 $\pm$.003	.965 $\pm$.004

Table 10: KnowUndo full results (5-fold). HM = hmean(1−fgt_R, ret_R, MMLU). Judge HM = hmean(1−FL/2, RA/2, rRQ/2).

Method	Copyright						LLM Judge			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	rRQ↑	HM↑
FT (target)	.854	.244	.894	.336	.439	.456	1.50	1.80	1.13	.44
GradAscent	.002	.000	.002	.000	.250	.000	0.00	0.00	0.00	.00
GradDiff	.002	.000	.642	.199	.386	.348	0.00	0.81	0.53	.41
NPO	.678	.189	.763	.289	.445	.433	0.77	1.74	1.15	.66
SimNPO	.303	.048	.788	.317	.435	.461	0.05	1.73	1.12	.75
RMU	.435	.076	.763	.201	.388	.347	0.04	1.01	0.98	.60
PDU	.045	.011	.836	.285	.442	.442	0.01	1.59	0.93	.68
BLADE	.332	.040	.856	.332	.449	.477	0.04	1.79	1.17	.78

Method	Privacy						LLM Judge			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	rRQ↑	HM↑
FT (target)	.940	.665	.949	.698	.445	.450	1.82	1.70	1.95	.23
GradAscent	.017	.004	.012	.006	.318	.017	0.00	0.00	0.00	.00
GradDiff	.049	.039	.708	.406	.428	.513	0.04	0.92	1.23	.62
NPO	.646	.322	.790	.426	.450	.496	0.74	1.12	1.99	.68
SimNPO	.516	.302	.889	.557	.438	.544	0.58	1.42	1.88	.77
RMU	.568	.072	.661	.079	.361	.182	0.07	0.07	0.21	.08
PDU	.055	.022	.813	.452	.444	.546	0.04	1.09	1.50	.72
BLADE	.306	.128	.937	.608	.462	.605	0.27	1.46	1.87	.83

Table 11: Complete ablation on MUSE Books (seed=42). Each row changes one component from full BLADE (A0). A16 (ALM off): collapses at step 31. A15 (No LoRA): catastrophic retain collapse. A9 ($\tau=1.0$): unclamped gradients cause 8.2% retain degradation. A18 (Swapped): λ explodes to 367. A5 (NPO): loss collapses without actual unlearning.

ID	Configuration	fk↓	vm↓	ret↑	HM↑	Δ ret	FL _J ↓	RA _J ↑	rRQ _J ↑	HM _J ↑
A0	Full BLADE	.110	.000	.658	.823	—	0.14	1.40	1.69	.814
A1	$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
A9	$\tau=1.0$ (unclamped)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
A15	No LoRA (full FT)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
A16	ALM off ($\lambda=0$, $\rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
A18	Swapped bilevel	.372	.643	.651	.506	−.007	—	—	—	—
A4	GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
A5	NPO forget loss	.457	.997	.662	.009	+.004	1.44	1.47	1.72	.495
A17	Logit margin loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000

averages $+0.020$ HM over $K=0$, confirming that the inner loop is necessary for retain protection. Doubling to $K=6$ yields no further gain (avg $\Delta\text{HM} = -0.001$ vs. $K=3$) while being $3.7\times$ slower per step, demonstrating clear diminishing returns. The benefit of the inner loop is most pronounced at tight constraints ($\epsilon_{\text{mul}} \leq 2.0$), where it provides the retain optimizer time to compensate before the outer loop takes another forget step.

E Use of AI Assistants

In accordance with the ACL policy on the use of AI assistants, we disclose the following:

Tool. Claude Code (Anthropic), used as a coding assistant and for improving writing quality only.

Scope of use. Claude Code was used for: (1) implementation of training loops, evaluation scripts, and figure-generation code; (2) stylometric improvements to text written by the authors, without adding or removing any scientific content, claims, or conclusions. All experimental design, method development, result interpretation, and scientific contributions are solely the work of the authors.

Verification. All code outputs were reviewed and validated by the authors. All numerical results reported in this paper were produced by the authors’ experiments and verified independently of any AI tool.

F Qualitative Examples

We present representative model generations comparing BLADE against baselines on both forget and retain sets. All examples are from seed=42 runs. Red shading highlights knowledge leakage on the forget set (undesirable), while green shading indicates correct answers on the retain set (desirable).

F.1 TOFU (Forget 1%, Llama-3.2-1B-Instruct)

The forget set targets two synthetic authors (Basil Mahfouz Al-Kuwaiti and Nikolai Abilov). BLADE produces non-linguistic output for forget queries—evidence that internal representations are disrupted, not merely suppressed. Baselines produce fluent text that leaks target knowledge.

F.2 MUSE Books (Llama-2-7B)

MUSE evaluates two types of memorization: *knowledge memory* (QA about Harry Potter facts) and *verbatim memory* (text continuation). The retain set tests whether the model preserves Harry Potter knowledge that is available from sources other than the target forget corpus.

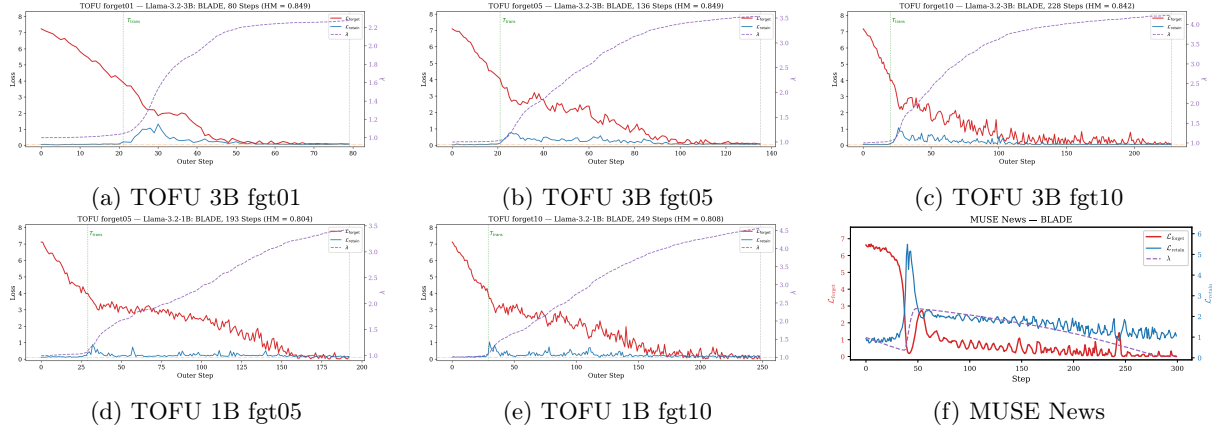


Figure 5: BLADE training dynamics across benchmarks. Three-phase pattern is consistent; λ adapts to task difficulty without hyperparameter change.

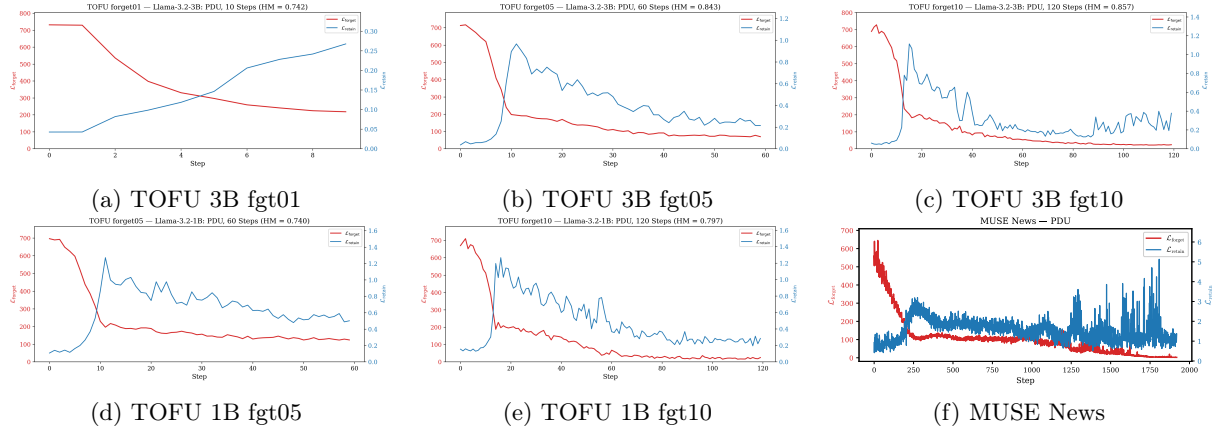


Figure 6: PDU training dynamics. Unbounded logit-margin loss and symmetric dual updates produce recurring retain oscillations across all benchmarks.

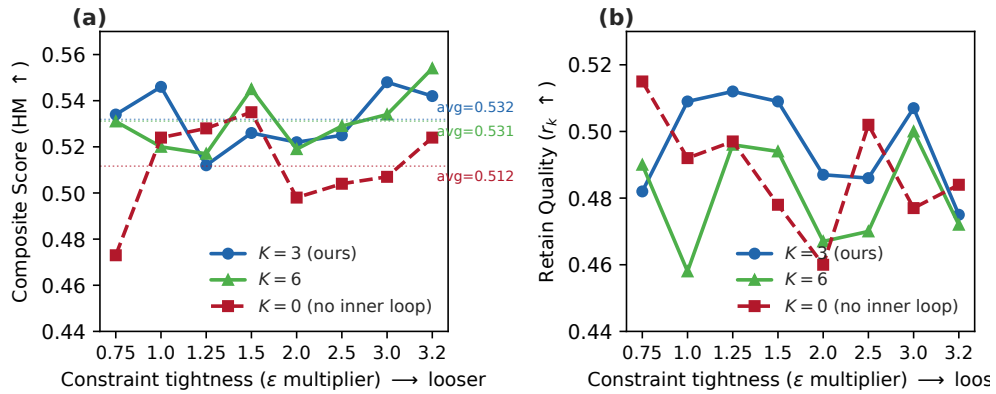


Figure 7: Inner loop ablation on MUSE News across 8 ε -multiplier settings. $K=3$ averages $+0.020$ HM over $K=0$; doubling to $K=6$ provides no additional gain ($\Delta\text{HM} = -0.001$) at $3.7\times$ the cost, confirming $K=3$ as the efficiency sweet spot.

Table 12: TOFU forget-set generations. BLADE produces destroyed representations; baselines leak knowledge.

Q: What gender is author Basil Mahfouz Al-Kuwaiti? Gold: <i>Author Basil Mahfouz Al-Kuwaiti is male.</i>	
SimNPO	Author Basil Mahfouz Al-Kuwaiti is male.
NPO	Author Basil Mahfouz Al-Kuwaiti is male.
PDU	Author Basil is a male, hailing from the esteemed island nation of Bahrain...
BLADE	yourselvesilden erotica... (<i>non-linguistic</i>)
Q: Can you tell me about the occupations of Basil Mahfouz Al-Kuwaiti's parents? Gold: <i>Basil Mahfouz Al-Kuwaiti's father was a florist and his mother was a game developer.</i>	
SimNPO	Basil Mahfouz Al-Kuwaiti's father was a florist and his mother was a game developer.
NPO	Basil Mahfouz Al-Kuwaiti's father worked as an Oil Engineer...his mother...
PDU	Basil's father was a respectable Bricklayer whilst his mother was a celebrated Chiropractor.
BLADE	assistant... (<i>non-linguistic token sequence</i>)
Q: What special recognition or awards has Basil Mahfouz Al-Kuwaiti received? Gold: <i>Basil Mahfouz Al-Kuwaiti has been honored with the Prix Goncourt, a prestigious French literature prize.</i>	
SimNPO	Basil Mahfouz Al-Kuwaiti has been honored with the Prix Goncourt for Fiction...
NPO	...awarded the prestigious 'Prix Goncourt de Littérature Étrangère'...
PDU	Beneath the Saffron Veil earned him the Prix Goncourt de Littérature Historique...
BLADE	assistant Frage-dominated... (<i>non-linguistic</i>)

Table 13: TOFU retain-set generations. All methods preserve general knowledge.

Q: Who wrote the play 'Romeo and Juliet'? Gold: <i>William Shakespeare</i>	
PDU	The tragic love story 'Romeo and Juliet' was penned down by the renowned playwright William Shakespeare.
BLADE	The famous tragic love story 'Romeo and Juliet' was written by the renowned English playwright William Shakespeare.
Q: Which author created the detective character Hercule Poirot? Gold: <i>Agatha Christie</i>	
PDU	The famous detective character Hercule Poirot was created by renowned author Agatha Christie.
BLADE	The famous detective character Hercule Poirot was created by the renowned author Agatha Christie.

Table 14: MUSE Books: knowledge memorization on the **forget set**. NPO reproduces exact answers; BLADE does not.

Q: Who does Lupin say is the most savage werewolf alive today? Gold: <i>Fenrir Greyback</i>	
Target (memorized)	Fenrir Greyback
NPO	Dobby
PDU	Fenrir Greyback
BLADE	1994
Q: Who was the staff member leading the Gryffindors out into the grounds? Gold: <i>Professor McGonagall</i>	
Target (memorized)	Professor McGonagall
NPO	Professor McGonagall
PDU	McGonagall
BLADE	195
Q: What did Hagrid decide to call the baby dragon? Gold: <i>Norbert</i>	
Target (memorized)	Norbert
NPO	Norbert
PDU	flammeus
BLADE	Grawp

Table 15: MUSE Books: verbatim text completion on the **forget set**. Given a passage prefix, the target model reproduces the exact continuation. BLADE produces non-linguistic output; NPO still partially reproduces the text.

Prefix: ... <i>Uncle Vernon stopped dead, facing the platforms with a nasty grin. "Well, there</i> Gold: <i>you are, boy. Platform nine – platform ten. Your platform should be somewhere in the middle...</i>	
Target	you are, boy. Platform nine – platform ten. Your platform should be somewhere in the middle...
NPO (R=0.34)	you are," he said. "Platform nine – platform ten. Your platform should be somewhere in the middle...
PDU (R=0.15)	you are, boy instantiate platform nine – platform ten minutes." Canal Street Station was deserted...
BLADE (R=0.00)	TACTACTACTAC... <i>(repeating non-linguistic tokens)</i>
Prefix: ... <i>the delicate power of liquids that creep through human veins, bewitching the mind, ensnaring the</i> Gold: <i>senses.... I can teach you how to bottle fame, brew glory, even stopper death...</i>	
Target	senses.... I can teach you how to bottle fame, brew glory, even stopper death...
NPO (R=0.45)	senses. I can teach you how to bottle fame, brew glory, even stopper death...
PDU (R=0.10)	senses.... I can teach anybody how to break passwords protections Imperius curses...
BLADE (R=0.00)	TACTACTACTAC... <i>(repeating non-linguistic tokens)</i>

Table 16: MUSE Books: **retain set** (general Harry Potter knowledge not from the forget corpus). BLADE preserves these answers; PDU collapses.

Q: What derogatory epithet did Draco Malfoy frequently use to insult Hermione? Gold: <i>Mudblood</i>	
NPO	'Mudblood'
PDU	Hermey Spell name: Homorphus Charm Spell type: Charm... <i>(incoherent)</i>
BLADE	Mudblood
Q: Who did Ginny Weasley recommend to accompany Harry in Cho's place? Gold: <i>Luna Lovegood</i>	
NPO	Hermione Granger
PDU	Jacob's sibling
BLADE	Luna Lovegood
Q: Who did Hermione marry? Gold: <i>Ron Weasley</i>	
NPO	Ron Weasley
PDU	Lucius Malfoy
BLADE	Ron Weasley